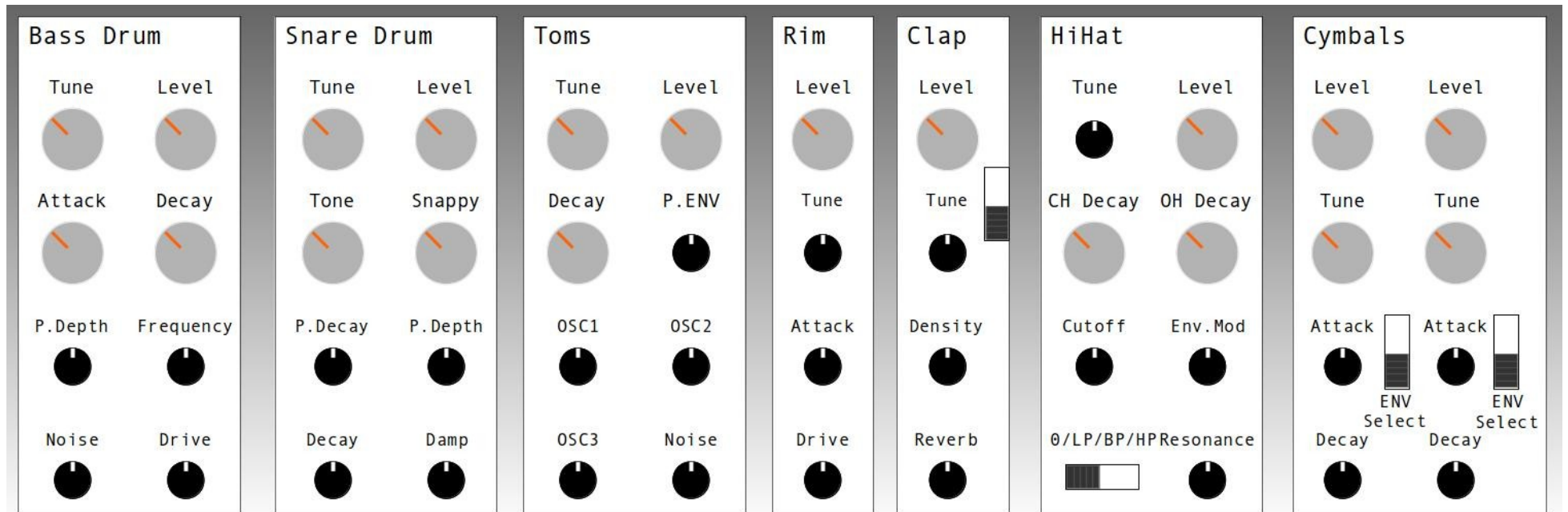


Extra9 - NAVA Expansion

by Wellental Schall Apparaturen



Introduction

Extra9 is a dedicated expansion for your NAVA - TR-909 clone by e-licktronic and is sold as a DIY electronic parts kit by Wellental Schall Apparaturen.

The expansion is meant to be installed on top of the existing NAVA main-board and fits seamlessly onto its layout.

With Extra9 you add a total of 37 extra pots and 4 extra switches to your NAVA TR-909 clone which increases the sonic potential of the popular TR-909 drum instruments tremendously.

Adding the Extra9 modifications means that you get more possibilities to manipulate the synthesis parameters of each instrument while you can always go back to the original character.

All the modifications have been thoroughly picked and tested for best practical sonic manipulations and of course all modifications are fully analog!

Disclaimer

Extra9 is a do-it-yourself electronic parts kit. Besides the expected functioning of the delivered components, Wellental Schall Apparaturen may not be made responsible for any damages you may cause to the components, the fully or partly assembled Extra9 expansion or your NAVA TR-909 clone.

All assembling and installation is at your own risk!

General Extra9 Assembling Instructions

Extra9 Kits / BOM

First run of Extra9 kits contain the following parts (as of May '18 - may change in later runs):

- Extra9 PCB -with all IC preinstalled
- plastic isolation sheet -CNC milled in the shape of the PCB
- all (37) pots needed -including 6 specific reverse-logarithmic pots that are not available from any seller
- all (4) switches needed
- all (13) connectors needed -with preinstalled wires / +2 options for power supply connection
- 2 matched transistors -for HiHat VCF
- 2 SMD transistors
- 1.5nF / 680nF film capacitors
- plastic spacers

There is also a limited run of full kits "Extra9 Kit+" available that contain all other parts needed:

- all SMD parts needed -resistors, capacitors, trimmers, diodes
- all thruhole parts needed -replacement resistors and capacitors for NAVA board

A User BOM with all parts not included in the Extra9 Kit is available as file to download. Also there are public shopping-carts for Reichelt and Mouser published for those parts. Shopping-carts and Extra9 Kit+ have 1-2 spare pieces on each SMD capacitor/resistor/diode. Those parts are very small and it may happen that a part gets lost during assembly, so.. those extra parts are your day-saver!

User BOM (pick your correct Version!):

<https://www.dropbox.com/sh/ygxkfbkcs7za5qt/AACI3d9s16Sr2VsEFgnJMatma?dl=0>

shopping-carts 1st Run/PCB V3.2

Reichelt shopping-cart 1st Run/PCB V3.2 (last checked 2018-05-22): <https://www.reichelt.de/my/1456952>

Mouser shopping-cart 1st Run/PCB V3.2 (last checked 2018-05-22): <https://www.mouser.ch/ProjectManager/ProjectDetail.aspx?AccessID=79abe0ae9d>

Shopping-carts 2nd Run/PCB V3.2.1

Reichelt shopping-cart 2nd Run/PCB V3.2.1 (last checked 2019-05-01): <https://www.reichelt.de/my/1581046>

Mouser shopping-cart 1st Run/PCB V3.2 (last checked 2019-05-01): <https://eu.mouser.com/ProjectManager/ProjectDetail.aspx?AccessID=bbec120381>

Tools needed

- Tweezers to place SMD parts (I prefer those with bent tips)
- Soldering iron with a small tip (I use 1,2mm flat top)
- Solder (small diameter & Pb recommended for hand soldering, because of lower soldering temperature - I use Sn62Pb36Ag2 0,50mm)

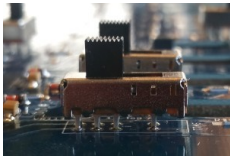
Soldering of SMD Parts

- Apply some solder to only one pad of a part.
- Use tweezers to position the part while heating the solder of that first pad.
- When correctly positioned let cool down the solder. Do not release the tweezer before solder has cooled.
- Now apply solder to the remaining pad

Soldering of the thruhole parts (pin headers, switches, pots)

It's important to maintain the bottom side of the board as flat as possible. Thus, thruhole parts are ideally soldered on component side!
If not possible, it's best to clip off component leads flat to the board before soldering.

- **Pin Headers** included with kits are painful to solder on top side. Put the headers in place and solder on pin on top side. Then cut the legs on bottom side flat to the board and solder all pins on bottom side.
- **Switches must not rest directly on PCB but need to be installed with some spacing to the board!** The upper edge of the body must be 7mm above the board. I highly recommend to solder switches after installation of the board to the housing for perfect alignment.
If you cant wait: legs are +/- of the right length already, solder one corner pin flush to the PCB bottom. Then check for positioning, flat leveling and in particular for 7mm spacing of the upper edge and correct them if needed. Only solder remaining pins when switches are nicely positioned.



- **Pots** should be soldered as last to avoid any damage to the shafts by the soldering iron. Go one by one from left to right (or vice versa if left-handed). Pins are soldered on component side. Cut pins flush on bottom side after soldering.
On some pots you may have to shorten plastic nipples on top of housing (particularly the green ones in your kit) - top edge of the pot housing must not exceed the PCB by more than 7mm!

Pot codings:

Variation 1	curve{A,B,C} - value(xx) - multiplier{K, M}	A:log, B: lin, C: rev-log	K: kilo-ohms, M: mega-ohms
Variation 2	curve{A,B,C} - value(xxy)		xx: value, y: multiplier 10 ^y (e.g. 105 = 1M0hms)

Pot indicator marking:

For indicator-line coloring I recommend using a gloss paint marker with extra-fine tip (~0.8mm..1mm)



Desoldering and replacing of NAVA board parts

For every section (instrument) there is a handful of parts that must be removed or replaced on the NAVA board (mainly resistors and capacitors). This is the most annoying part of the Extra9 installation procedure. Replacement of some parts (resistors) can be avoided by simply soldering a second part in parallel, which is recommended. Both options are included in BOM/Kit.

General advice: Take your time! – pads are very sensitive to excessive force – they may come off if you pull to hard or don't let solder melt entirely!

Best practice:

- Cut one of the legs.
- Heat the second leg and gently remove the part with a pliers. If part does not come off easily add more solder to that leg.
- Use a desoldering pump to remove solder from pads, later you will have to solder wires to most of the pads. (Heating from one side while sucking from the other side is easiest!)

Connecting the wires on NAVA PCB

Wires are connected on bottom/component side of NAVA PCB. Most of the wires are connected to previously desoldered pads, some wires have to be soldered directly to a part in place. For pads, solder can be applied either on component side or top side.

I recommend to cut the wires to a convenient length. Therefore you should install the Extra9 board as well as the connectors and run the wire to its intended position. Give some extra centimeters for easy unplugging/plugging.

Installation of the finished Extra9 board into the NAVA

Space between NAVA mainboard and housing is very limited (10mm). Extra9 pots take up most of this space (7mm). Further we have to include the Extra9 PCB which is minimised to 1mm and a sheet for electrical isolation between NAVA and Extra9 boards. I choosed to have an isolation sheet of again 1mm to ensure proper separation. If we sum this up, we see that there is another 1mm left for solder-points on top side of NAVA PCB and bottom side of Extra9 PCB. So this is the reason why I choosed SMD parts and recommend to solder thru-hole parts on top side of Extra9 PCB.

- First step in installation process is to shorten the solder points/pins on the NAVA PCB which sit under the Extra9 board as flat as possible to the board, max. 1mm! Use an angled side cutter!
- Place the plastic spacers on the top 6 holes of the NAVA board. Gluing them down is recommended.
- Place the plastic isolation sheet under the Extra9 board and stick both together by using some pieces of tape at the edges.
- Now place the Extra9 PCB onto the NAVA mainboard and make all the connections.
- Finally place the whole package into the case at once and fix it with the screws.



Installation of the finished Extra9 board into a original NAVA case

On original NAVA cases all spacers are of the same length (10mm). To install the Extra9 modification you either shorten the 6 upper standoffs by 3mm or enlarge the installation holes in the Extra9 PCB as well as the plastic sheet and spacers to the exact diameter of your standoffs (you could also leave out the spacers btw.) I recommend the later. Since everything is packed very tightly between NAVA PCB and case, nothing will move again.

- Of course you have drill the new holes to your NAVA case first.
- Use a step drill to make the holes of 8mm!
- For the square holes make a small hole in the middle (3mm) and then work out the holes with a fine file.
- Be patient doing this works!



Drill template can be downloaded. It's split in 2 parts to fit it on a A4 paper. Print it on a foil or thin paper. Check dimensions!

PDF A4 https://www.dropbox.com/s/6otd8p2l6xr602z/Extra9_NAVA_Drilling_A4.pdf?dl=0

SVG https://www.dropbox.com/s/plly0lzrh53lgab/Extra9_NAVA_Drilling.svg?dl=0

Assembling the Extra9 PCB & Make the connections to the NAVA PCB

Now lets start assembling!

We go section by section. Finish soldering all the Extra9 sections before you start making the connections to the NAVA board. Read each section carefully before getting to work. On some of the instruments there are optional modifications which you may not want to include or which may be realised on the NAVA board only.

Some instructions on reading the Installation Manual

Connection points on NAVA board

Every instrument section in this manual has a drawing of the NAVA board component side. Boxes around some parts and additional pointers indicate connection points that are connected by wires to the pin headers of the Extra9 board.

The boxes are in different colors:



Red Boxes indicate parts on the NAVA PCB that have to be removed for Extra9 connections.



Blue Boxes indicate parts on the NAVA PCB that should be replaced by a different value. Either by replacement of the part or by soldering a second part in parallel (only resistors). Parts for both options are listed in BOM.



Purple Boxes indicate parts on the NAVA PCB that are connection points for Extra9 but must not be removed.

Pin headers on Extra9 board

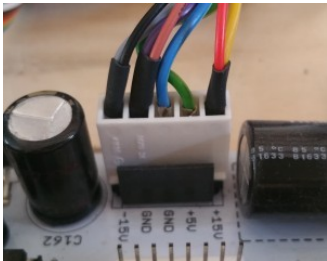
Pin header pins are numbered from left to right on the Expansion Board. Pin 1 has a rectangular pad other pins are oval.

Power Supply

Extra 9 needs +15V / GND / -15V

We use the supply voltages at connector J8 on the NAVA IO Board which also is used to supply the NAVA main board.

If you use crimped connectors like on the NAVA BOM you could use new crimp terminals and include wires for the Extra9 power connector at respective pins. Use a piece of shrinking tube to tighten wires of a pin.



Otherwise I recommend to use a IDC (insulation displacement contact) connector like MTA-156.



Both options are included in Extra9 BOM or your Extra9 Kit.

BassDrum Components

J0_1 1x3 pin header

Start with the pin header if you prefer to solder it on top side!

C0_1 100u (elko)

C0_2 100u (elko)

C0_3 100u (elko)

C0_4 100u (elko)

R0_1 100R

R0_2 100R

R0_1 and R0_2 are not needed!

BassDrum

4 additional Pots

Tune Depth	Sets the mod amount of the pitch envelope. While the original 'Tune' pot basically sets the decay time.
Initial Tune	Sets the base frequency of the BD sine oscillator.
Noise	Basically a decay pot for the noise portion in the attack phase. Level of both noise and click is defined by Attack pot.
Drive	Overdrive of the BD sine oscillator.

Other Modifications

Attack Range	The expansion module includes a circuit which allows a much wider range of the attack level set by the Attack pot. The Attack portion of the BD is composed of a short click and a noise peak.
Decay Range	Decay range can be increased by changing C8 on NAVA board from 0,33 to 1,0uF.
P.Decay Range	Pitch decay range (set by the Tune pot) can be extended by changing C9 on NAVA board from 0,33 to 1,0uF. To maintain the minimum values of the pitch envelope R21 and R57 have to be lowered to a third of their original value, 750R and 7k5 respectively.

BassDrum Components

C1	100n (ceramic)
C2	100n (ceramic)
C3	33n (ceramic)

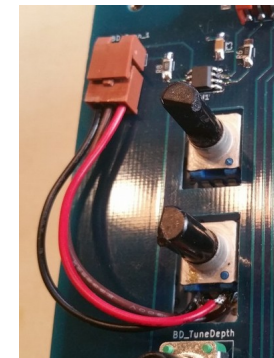
R1	6k8
R2_1	2M2
R2_2	2M2
R3	100k
R4	100k
R5	33k
R6	220k
R7	1k
R8	100R

U1	TL072
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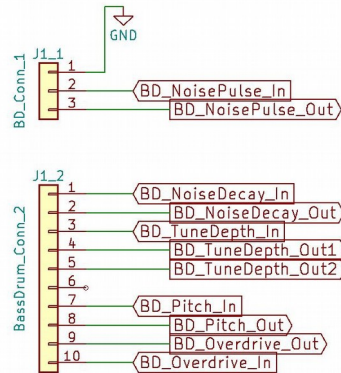
Install pots only after finishing all sections!

VR1	10k lin (B10k)
VR2	50k rev log (C50k)
VR3	1M lin (B1M)
VR4	10k lin (B10k)

J0_1	1x3 pin header facing downside (see pic)
J1_2	1x10 pin header



BassDrum - Connections



NoisePulse (Attack Range Increase)

Connections are made directly to Attack pot (VR3) pins and solder points! On top layer of NAVA board! Unsolder and bend up Pin 1 and 3 of Attack Pot (see Pic in section above)

- [1] - GND: Pin 1 on Pot
- [2] - NoisePulse_In : Pin 3 on PCB
- [3] - NoisePulse_Out: Pin 3 on Pot

NoiseDecay (variable Noise Decay)

remove R42

- [1] - NoiseDecay_In : R42 (junction R41)
- [2] - NoiseDecay_Out : R42 (junction Q6)

TuneDepth (variable TuneENV Depth)

remove R27, R29, R31

- [3] - TuneDepth_In : R29 (junction IC12A)
- [4] - TuneDepth_Out1 : R29 (junction IC12B)
- [5] - TuneDepth_Out2 : R31 (junction IC12B)

Pitch (variable Base Tuning)

remove R59

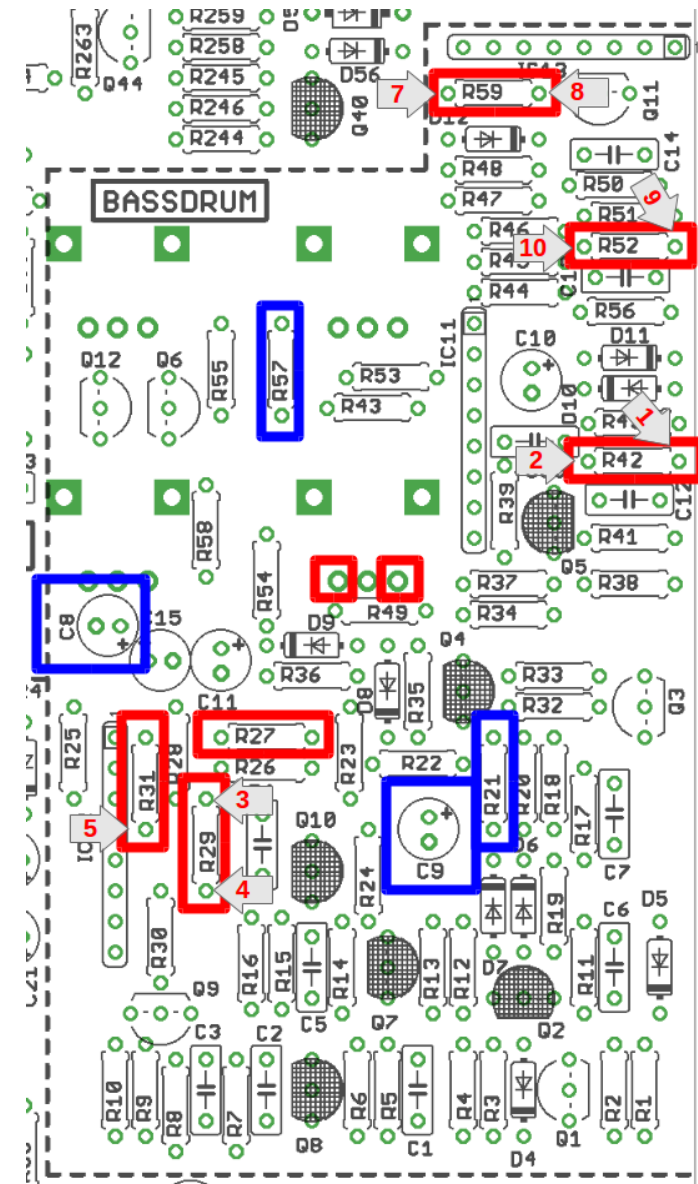
- [7] - Pitch_In : R59 (junction IC 12B)
- [8] - Pitch_Out : R59 (junction IC 13A)

Overdrive (variable OSC Overdrive)

remove R52

- [9] Overdrive_Out : R52 (junction D11)
- [10] Overdrive_In : R52 (junction IC13A)

NAVA Board V1.0



NAVA Board V1.02

Extended DECAY Range

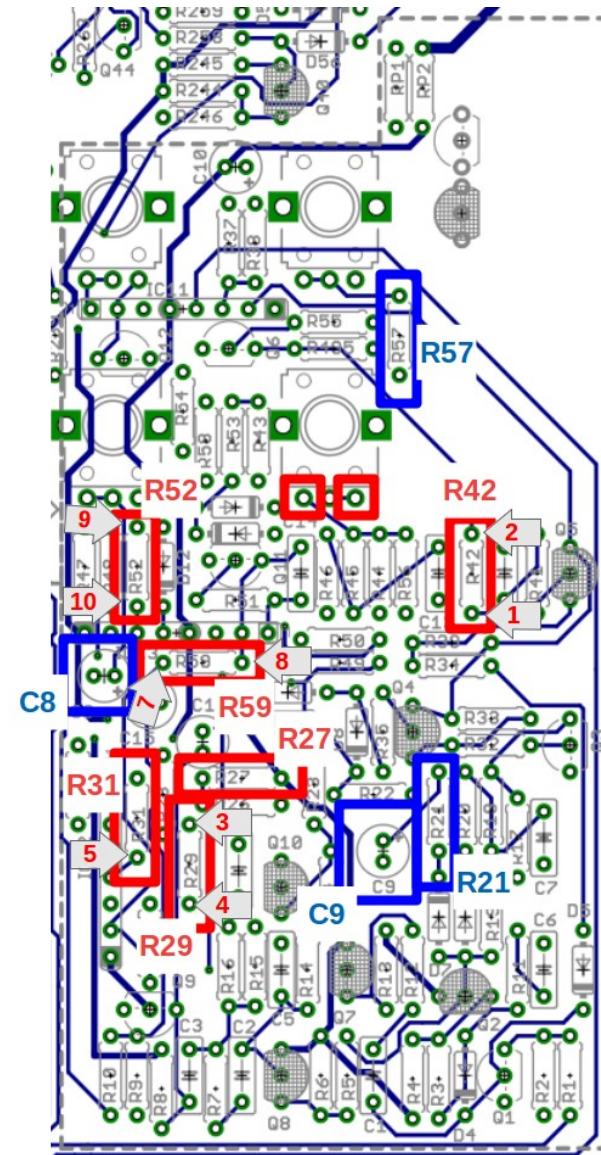
C8 : 1uF replace

Extended PITCH DECAY Range = TUNE Pot

C9 : 1uF replace

R21 : 750R replace or 1k parallel

R57 : 7k5 replace or 10k parallel



SnareDrum		
4 additional Pots		SnareDrum Components C11 100n (ceramic) C12 100n (ceramic) C13 100n (ceramic) C14 560n (ceramic) D10 1N4148 D11 1N4148 R10 22k R11 33k R13 220k R18 47k R19 1k5 U2 TL072 VR15 500k lin trimmer Install pots only after finishing all sections! VR11 1M lin (B1M) VR12 10k lin (B10k) VR13 1M log (A1M) VR14 1M rev log (C1M) J2 1x8 pin header
Pitch Decay	Sets the decay time of the pitch envelope.	
Pitch Amount	Sets the mod amount of the pitch envelope.	
Decay	Sets the decay time of the oscillator amp.	
Noise Damp	Basically a low pass filter for the noise portion.	
Other Modifications		
Tune Range	Due to the nature of the tuning circuit, the modification on the pitch envelope slightly shifts the range of the tune pot. To obtain a wider tune range, R284/R287 are reduced.	
Tone Range	Tone range (basically decay of the noise envelope) can be extended at the lower end by reducing R254.	
OSC2 Level	optional! The added Decay pot only affects one of two SD oscillators, OSC1. At slow decay values OSC1 appears much louder than OSC2, which is then almost inaudible. The expansion includes a simple mod which allows to increase OSC2 level to a more balanced/natural mix. The change is mainly noticeable at slow decay values but only slightly at shorts. Use Trimmer VR15 to find your preferred spot.	

SnareDrum - Connections



OSC1_ENV (variable Decay)

[1] - OSC1_ENV_In : D68 (junction C83)

PitchCV (variable Pitch Decay & Depth)

remove R248

R247 : 4k7 replace or 4k7 parallel

[2] - Trig In : R248 (junction R247)

[3] - PitchCV_Out : R248 (junction R249)

Noise_LPF (variable Noise Damp)

remove R300

[7] - Noise_LPF_In : R300 (junction R301)

[8] - Noise_LPF_Out : R300 (junction IC40A)

OSC2_ENV_Level (optional Level adjustment of OSC2)

[4] - OSC2_ENV_Level_In : R293 (junction R295)

[5] - OSC2_ENV_Level_Out : R293 (junction Q51)

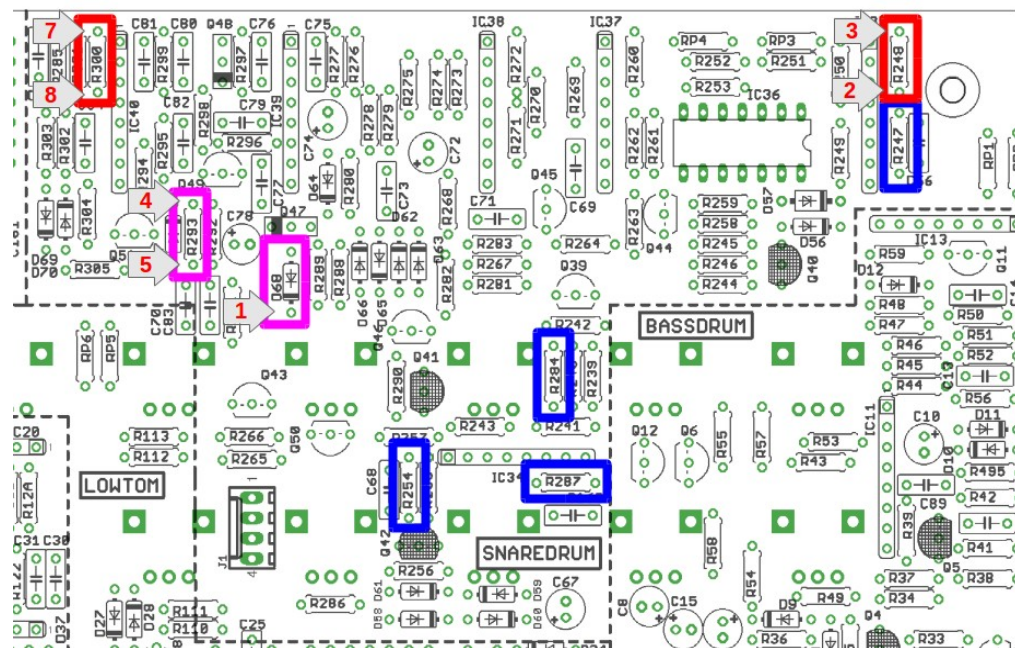
Adjust Tune Range

R284 : 3k replace or 4k7 parallel

R287 : 30k replace or 47k parallel

Extended TONE Range (Noise Decay)

R254 : 10k replace or 10k parallel



LowTom

5 additional Pots

Pitch Decay	Sets the decay time of the pitch envelope.
OSC1	Sets the level of Oscillator 1 (main drumbeat).
OSC2	Sets the level of Oscillator 2 (body).
OSC3	Sets the level of Oscillator 3 (attack peak).
Noise	Sets the level of the Noise portion on attack.

Other Modifications

OSC1 burst	optional! Oscillator has a short overdrive peak on the sine wave with every trigger. If you don't like this particular sound, you could just remove R102.
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LowTom Components

C21	100n (ceramic)
C22	100n (ceramic)
C23	47n (ceramic)

R20	33k
R21	470k
R22	2k2
R23	2k2
R24	2k2
R25	1k
R26	2k2
R27	2k2
R28	10k

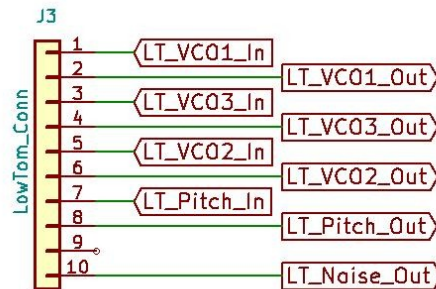
U3_1	TL072
U3_2	TL072

Install pots only after finishing all sections!

VR31	1M lin (B1M)
VR32	2k lin (B2k)
VR33	2k lin (B2k)
VR34	2k lin (B2k)
VR35	10k lin (B10k)

J3	1x10 pin header
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LowTom - Connections



VC01 (variable VC01 Level)

remove R104

- [1] - VC01_In : R104 (junction D21)
- [2] - VC01_Out : R104 (junction R99)

VC03 (variable VC03 Level)

remove R116

- [3] - VC03_In : R116 (junction D27)
- [4] - VC03_Out : R116 (junction R115)

VC02 (variable VC02 Level)

remove R106

- [5] - VC02_In : R106 (junction D23)
- [6] - VC02_Out : R106 (junction R98)

Pitch (variable Pitch Decay)

remove R61

- [7] - Pitch_In : R61 (junction D14)
- [8] - Pitch_Out : R61 (junction IC14B)

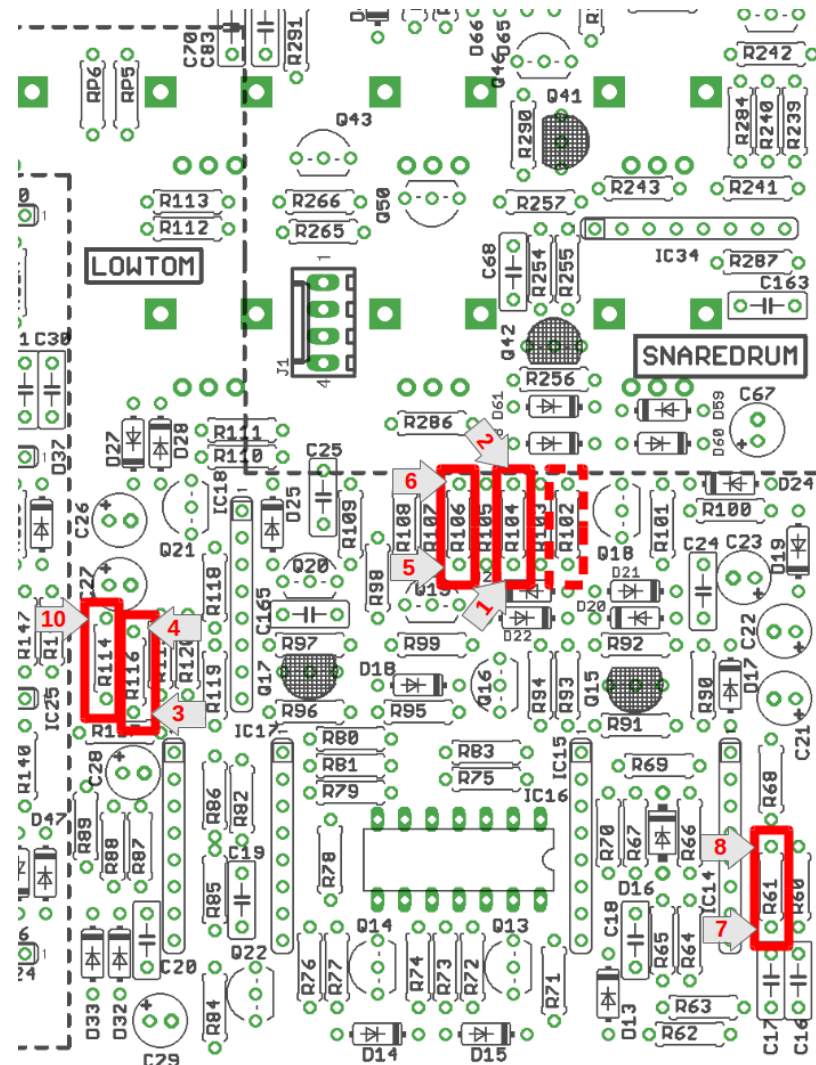
Noise (variable Noise Level)

remove R114

- [10] Noise_Out : R114 (junction R115)

optional: No burst of OSC1

remove R102



MidTom

5 additional Pots

Pitch Decay	Sets the decay time of the pitch envelope.
OSC1	Sets the level of Oscillator 1 (main drumbeat).
OSC2	Sets the level of Oscillator 2 (body).
OSC3	Sets the level of Oscillator 3 (attack peak).
Noise	Sets the level of the Noise portion on attack.

Other Modifications

OSC1 burst	optional! Oscillator has a short overdrive peak on the sine wave with every trigger. If you don't like this particular sound, you could just remove R163.
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MidTom Components

C31	100n (ceramic)
C32	100n (ceramic)
C33	47n (ceramic)

R30	33k
R31	470k
R32	2k2
R33	2k2
R34	2k2
R35	1k
R36	2k2
R37	2k2
R38	10k

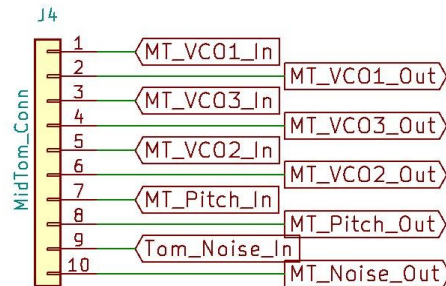
U4_1	TL072
U4_2	TL072

Install pots only after finishing all sections!

VR41	1M lin (B1M)
VR42	2k lin (B2k)
VR43	2k lin (B2k)
VR44	2k lin (B2k)
VR45	10k lin (B10k)

J4	1x10 pin header
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MidTom - Connections



VC01 (variable VC01 Level)

remove R165

[1]- VC01_In : R165 (junction D39)

[2] - VC01_Out : R165 (junction R159)

VC03 (variable VC03 Level)

remove R176

[3] - VC03_In : R176 (junction D46)

[4] - VC03_Out : R176 (junction R175)

VC02 (variable VC02 Level)

remove R167

[5] - VC02_In : R167 (junction D41)

[6] - VC02_Out : R167 (junction R160)

Pitch (variable Pitch Decay)

remove R122

[7] - Pitch_In : R122 (junction D36)

[8] - Pitch_Out : R122 (junction IC20B)

Noise (variable Noise Level)

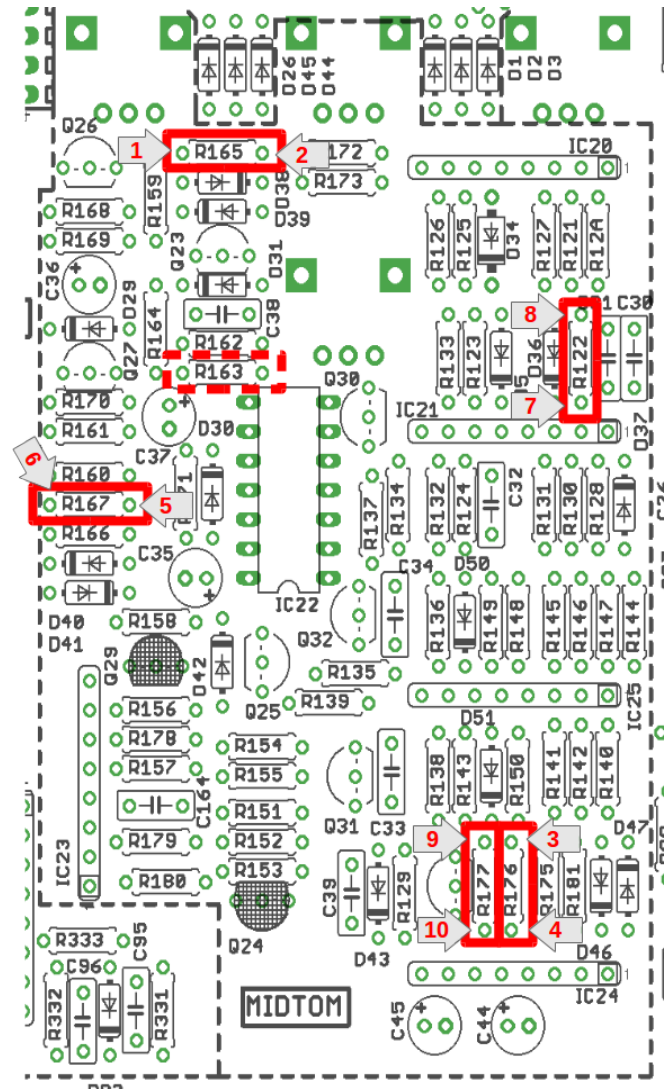
remove R177

[9] - Noise_In : R177

[10] - Noise_Out : R177 (junction R175)

optional: No burst of OSC1

remove R163



HiTom

5 additional Pots

Pitch Decay	Sets the decay time of the pitch envelope.
OSC1	Sets the level of Oscillator 1 (main drumbeat).
OSC2	Sets the level of Oscillator 2 (body).
OSC3	Sets the level of Oscillator 3 (attack peak).
Noise	Sets the level of the Noise portion on attack.

Other Modifications

OSC1 burst	optional! Oscillator has a short overdrive peak on the sine wave with every trigger. If you don't like this particular sound, you could just remove R359.
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HiTom Components

C41	100n (ceramic)
C42	100n (ceramic)
C43	47n (ceramic)

R40	33k
R41	470k
R42	2k2
R43	2k2
R44	2k2
R45	1k
R46	2k2
R47	2k2
R48	10k

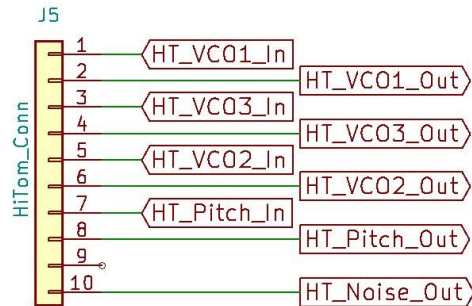
U5_1	TL072
U5_2	TL072

Install pots only after finishing all sections!

VR51	1M lin (B1M)
VR52	2k lin (B2k)
VR53	2k lin (B2k)
VR54	2k lin (B2k)
VR55	10k lin (B10k)

J5	1x10 pin header
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HiTom - Connections



VC01 (variable VC01 Level)

remove R357

- [1] - VC01_In : R357 (junction D78)
- [2] - VC01_Out : R357 (junction R351)

VC03 (variable VC03 Level)

remove R380

- [3] - VC03_In : R380 (junction D85)
- [4] - VC03_Out : R380 (junction R379)

VC02 (variable VC02 Level)

remove R355

- [5] - VC02_In : R355 (junction D80)
- [6] - VC02_Out : R355 (junction R350)

Pitch (variable Pitch Decay)

remove R332

- [7] - Pitch_In : R332 (junction D82)
- [8] - Pitch_Out : R332 (junction IC42B)

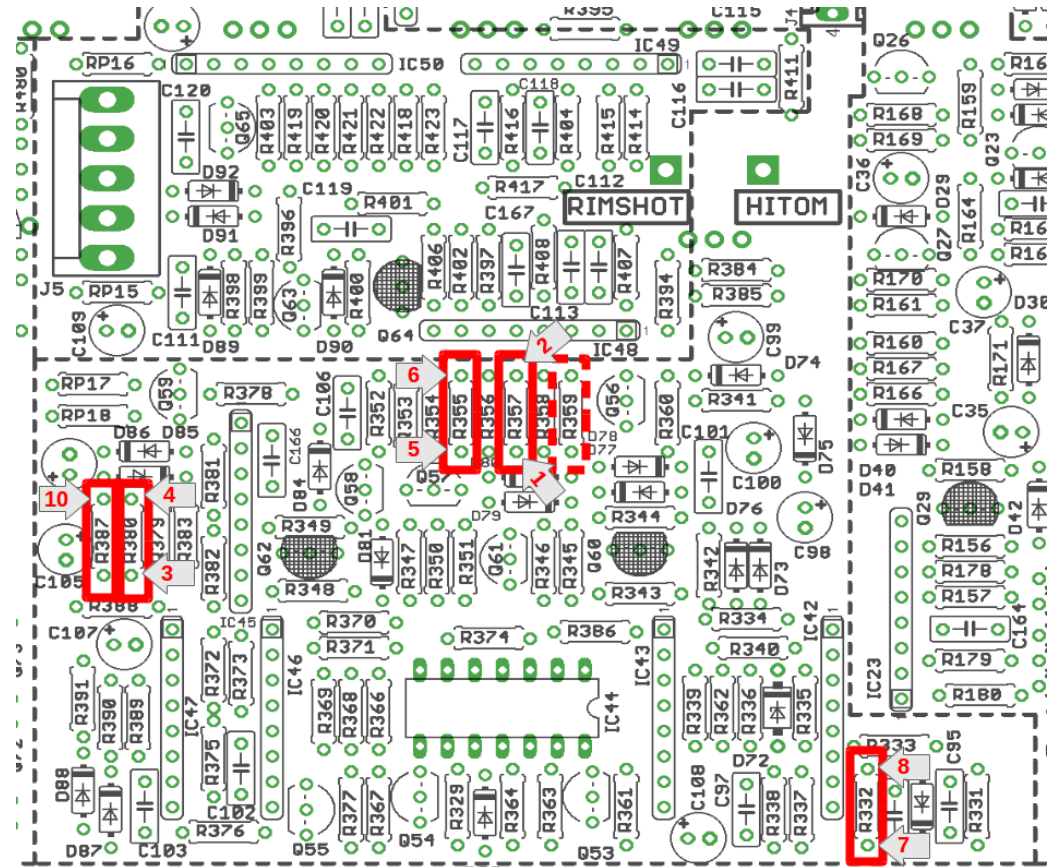
Noise (variable Noise Level)

remove R387

- [10] - Noise_Out : R387 (junction R379)

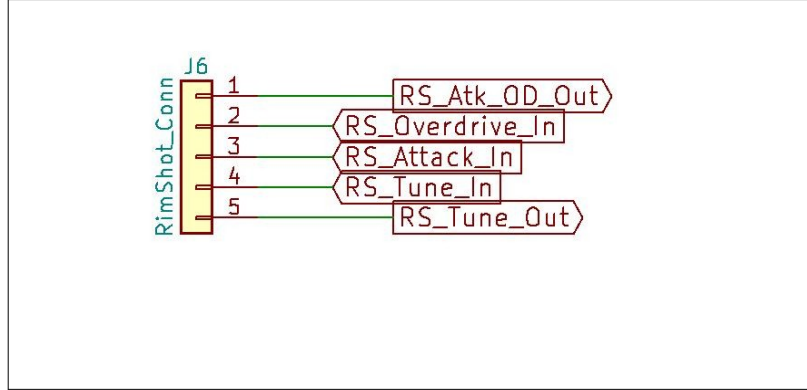
optional: No burst of OSC1

remove R359



RimShot		
3 additional Pots		RimShot Components
Tune	Sets the tuning of the main oscillator.	R50 100k
Attack	Sets the level of the attack oscillator.	R51 3k3
Drive	Overdrives the main oscillator.	R52 2k
Other Modifications		Install pots only after finishing all sections!
Level Range	optional! RimShot level is a little low compared to the other instruments. Level range can be increased by lowering R419.	VR61 1M rev log (C1M)
		VR62 10k lin (B10k)
		VR63 10k lin (B10k)
		J6 1x5 pin header

RimShot - Connections



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remove R417
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[1] - Atk_OD_Out : R417 (junction R408)

[2] - Overdrive_In : R415 (junction IC49A)

[3] - Attack_In: R417 (junction IC49B inv In)

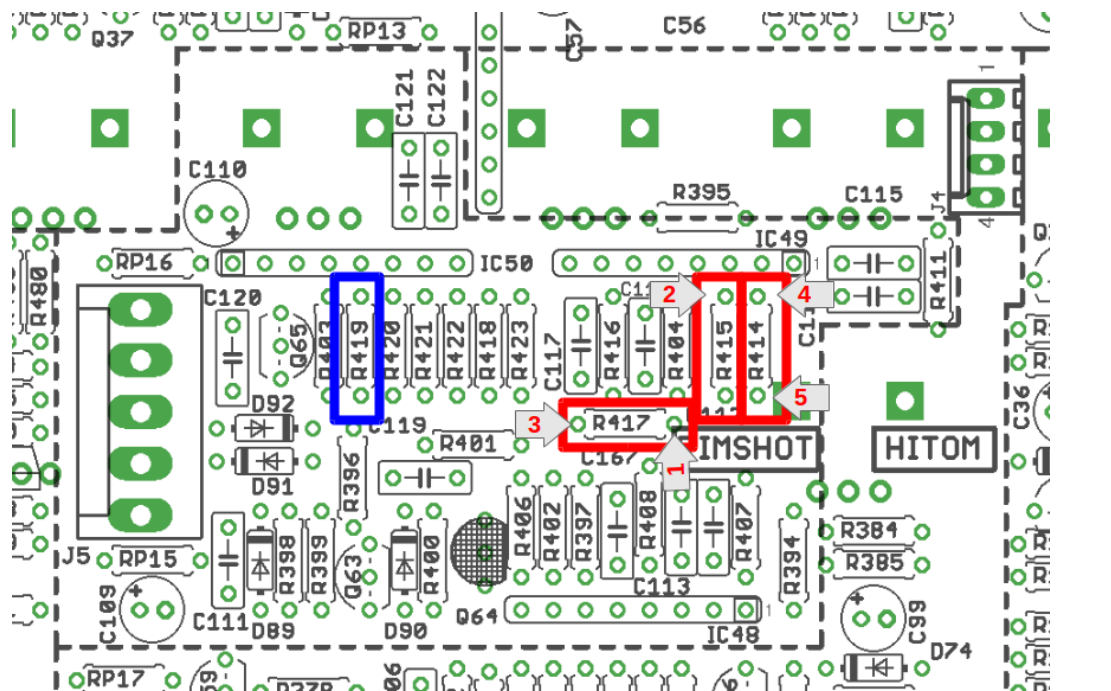
remove R414

[4] - Tune_In : R414 (junction IC49A inv In)

[5] - Tune_Out : R414 (junction IC49A Out)

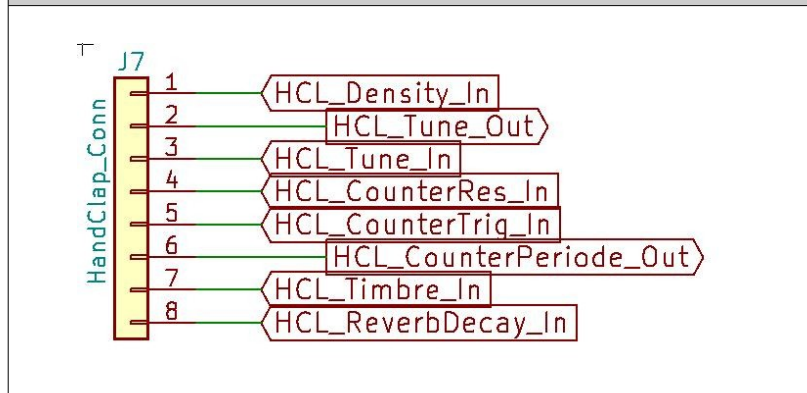
R419 : 4k7 parallel / 2k2 replace

(was 2k2 / 1k5 on V3.2 PCB)



HandClap		
3 additional Pots & 1 additional Switch		HandClap Components
Tune	Sets the tuning of the HandClap.	C60 330n (ceramic)
Timbre	Changes the tuning range and overall timbre of the HandClap sound.	C61 1n (ceramic)
Density	Sets the time interval between the HandClap peaks.	R60 10k
Reverb	Sets the decay time of the 'reverb' portion.	R61_1 10k
		R61_2 2k2
		R61_3 470R
		R62 100k
		R63 100k
		R64 10k
		R65 1k
		R66 2k2
		U7 CD4520
		SW7 SW_DP4T
		Install pots only after finishing all sections!
		VR71 1M rev log (C1M)
		VR72 500k log (A500k)
		VR73 1M lin (B1M)
		J7 1x8 pin header

HandClap - Connections



Tune (variable Tuning)

remove R207	
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```
[3] - Tune_In : R207 (junction IC26B inv In)
```

```
[2] - Tune_Out : R207 (junction IC26B Out)
```

Density (variable Clap Period)

remove C62	
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remove C64	
------------	--

remove R220	
-------------	--

R221 : 47k replace

C65 : 220nF (elko/tantalum) replace

```
[1] - Density_In : R220 (junction IC29 [1])
```

[4] - CounterRes_In : R213 (junction Q36)

```
[5] - CounterTrig_In : R221 (junction IC29 [13])
```

[6] - CounterPeriode_Out : C62 (junction IC29 [2])

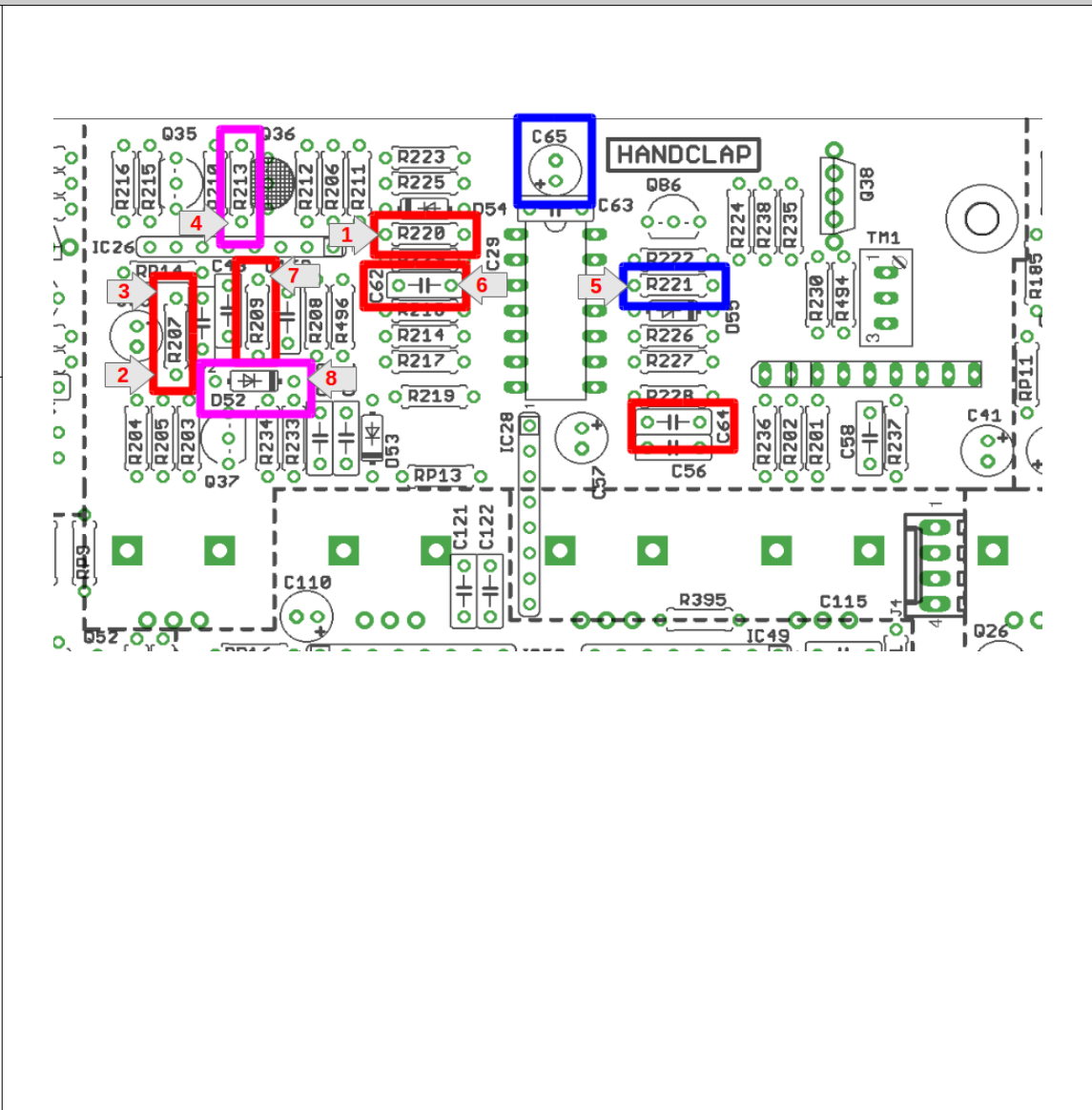
Timbre (variable Style)

remove R209	
-------------	--

[7] - Timbre_In : R209 (junction R208)

ReverbDecay (variable Reverb Decay)

[8] - ReverbDecay_In : connect to D52 (junction C59)



HiHats

HiHats have received a fully featured multimode filter with lowpass, bandpass and highpass options. The filter itself is based on the famous Steiner VCF (Steiner-Parker Synthacon). It's a diode filter that has a very distinctive sound which suits very well with the harsh NAVA/909 HiHats.

The circuit was adopted from the improved design of Yusynth.* Yves Usson has kindly authorized the use of his design in this project – so big big thanks to Yves... also for making his work available to the DIY community! I learned so much from his designs.

4 additional Pots & 1 additional Switch

Tune	Sets the tuning of the HiHats
Cutoff	Sets the cutoff frequency of the filter.
Envelope Mod	Sets the amount of cutoff modulation by the HiHats amp envelope. Envelope Mod is bipolar! In middle position no modulation is applied - negative modulation in counter clockwise position and positive modulation in clockwise position.
Resonance	Sets the amount of resonance on the filter.
Filter Type	Choose between off – lowpass – bandpass – highpass filter.

Other Modifications

CV In	optional! The expansion board hosts a second control voltage input that can be used for what ever you like (internal connection or external plug). Pin1 is the input, pin2 is GND. The input expects 5Vpp, if you apply 10Vpp you should replace R81_5 by 470k. An additional attenuation pot is recommended (pin1=plug, pin2=CV-In, pin3=GND).
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* <http://yusynth.net/Modular/EN/STEINERVCF/index-v2.html>

HiHat Components

J8_1 1x2 pin header
J8_2 1x3 pin header
J8_3 1x2 pin header

Start with the pin headers if you prefer to solder them on the top side! J8_3 is optional!

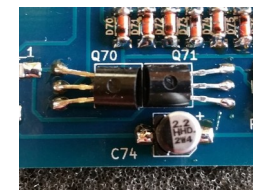
C77 1.5n (film)
C78 1.5n (film)
C79 1.5n (film)
C84 10n (film)
C71 100n (ceramic)
C71_2 100n (ceramic)
C72 100n (ceramic)
C72_2 100n (ceramic)
C81 680n (film)
C73 1u (elko)
C74 2u2 (elko)
C76 2u2 (elko)
C82 10u (elko)
C75 47u (elko)
C80 47u (elko)
C83 47u (elko)

D70 1N4148
D71 1N4148
D72 1N4148
D73 1N4148
D74 1N4148
D75 1N4148
D76 1N4148
D77 1N4148
D78 1N4148
D79 1N4148

Q70 BC547C
Q71 BC547C

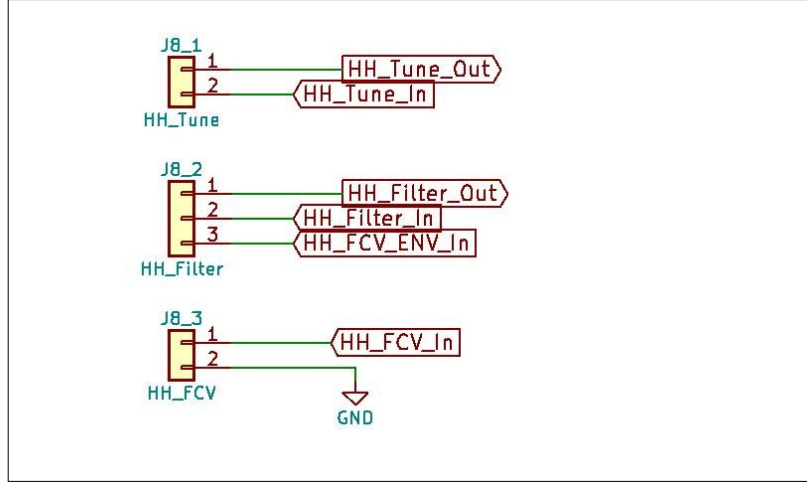
matched transistors – bend bodies down to PCB – flat side faces down to PCB – bodies should touch at the top

R82_2 82R
R75 390R
R76 1k
R80 1k
R85_3 1k



R81_4	1k2
R77	1k8
R78	1k8
R70	2k7
R82_1	3k9
R84	3k9 (was 18k on V3.2 PCB)
R71	10k
R81_6	10k
R86_1	10k
R87_3	10k
R84	18k
R87_4	20k
R83_3	22k
R72	47k
R73	47k
R83_2	47k
R86_2	47k
R81_1	68k
R87_1	100k
R87_2	100k
R87_6	100k
R87_7	100k
R81_3	150k
R87_5	150k (was 220k on V3.2 PCB)
R74	180k
R79	180k
R81_5	220k
R85_1	220k
R81_2	470k
R85_2	1M
R83_1	2M2
U8	TL074
U9	TL074
SW8	SW_DP4T
VR80	1k lin (trimmer)
VR85	2k lin (trimmer)
VR86	50k lin (trimmer) (not on V3.2 PCB)
Install pots only after finishing all sections!	
VR81	50k rev log (C50k)
VR82	50k lin (B50k)
VR83	50k rev log (C50k)
VR84	50k lin (B50k)

HiHat - Connections



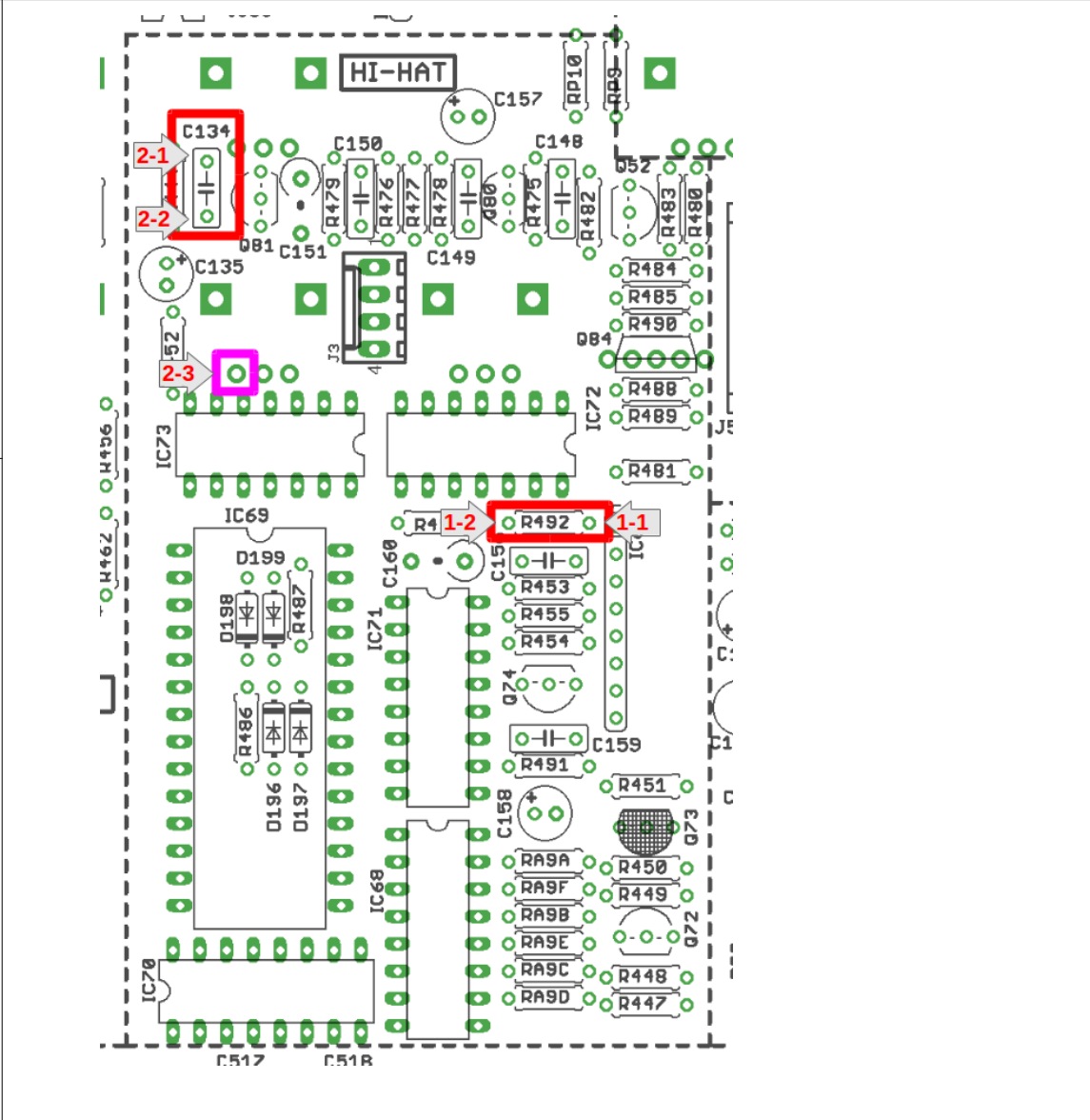
remove R492

```
[1-1] - Tune_Out : R492 (junction IC72c)
[1-2] - Tune_In  : R492 (junction R493)
```

```
remove C134
```

```
[2-1] - Filter_Out : C134 (junction R444)
[2-2] - Filter_In : C134 (junction R478)
[2-3] - FCV_ENV_In : Pin3 of either CH/OH DecayPot
```

connect Additional Control Voltage



Cymbals

Both Cymbals Ride/Crash have receive an alternative envelope with variable attack and decay controls. Both controls have a range of 3ms...1,3s. Maximum time of 1,3s matches the length of the cymbal samples at the lowest tuning. Hence, at higher tuning max time is longer than the actual sample length. Likewise the envelope will become much longer than the actual sample if both controls are turned to high values.

It's important to keep this in mind! The original envelope is derived from the eprom address counter and thus always matches the length of the sample, regardless of the tuning.

A switch lets you choose between original/alternative envelope. Furthermore the alternative envelope can operate in non-retriggered/retriggered modes. The modes are mainly of interest at slow attack times. If retriggered, the envelope will always start from 0 going thru full attack phase. In non-retriggered mode the envelope starts the attack phase at the current level of the envelope. E.g. if decay phase of the last note has only finished by let's say 50%, as a new trigger appears the envelope will start at that level and attack phase will be shortened (in this case to 50% of the value set by the pot).

Both Cymbals: 2 additional Pots & 1 additional Switch

Attack	Sets the attack time of the alternative envelope.
Decay	Sets the decay time of the alternative envelope.
Mode	Choose the envelope mode: original envelope – alternative AD envelope non_retriggered – alternative AD envelope retriggered

Cymbals Components

C94	10n (ceramic)
C95	10n (ceramic)
C104	10n (ceramic)
C105	10n (ceramic)
C91	100n (ceramic)
C92	100n (ceramic)
C101	100n (ceramic)
C102	100n (ceramic)
C96	1u (tantalum or ceramic)
C106	1u (tantalum or ceramic)

D90	1N4148
D91	1N4148
D92	1N4148
D100	1N4148
D101	1N4148
D102	1N4148

Q90	MMBT3904
Q100	MMBT3904

R94	150R
R104	150R
R93	3k9
R103	3k9
R92	10k
R102	10k
R99_2	15k
R109_2	15k
R99_1	33k
R109_1	33k
R90	100k
R91	100k
R97	100k
R100	100k
R101	100k
R107	100k
R95	200k
R96	200k
R105	200k
R106	200k
R98_1	270k
R98_2	270k
R108_1	470k
R108_2	470k

U10	TLC555
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U11	TL074
U12	TLC555
U13	TL074

SW10	SW_DP3T
SW11	SW_DP3T

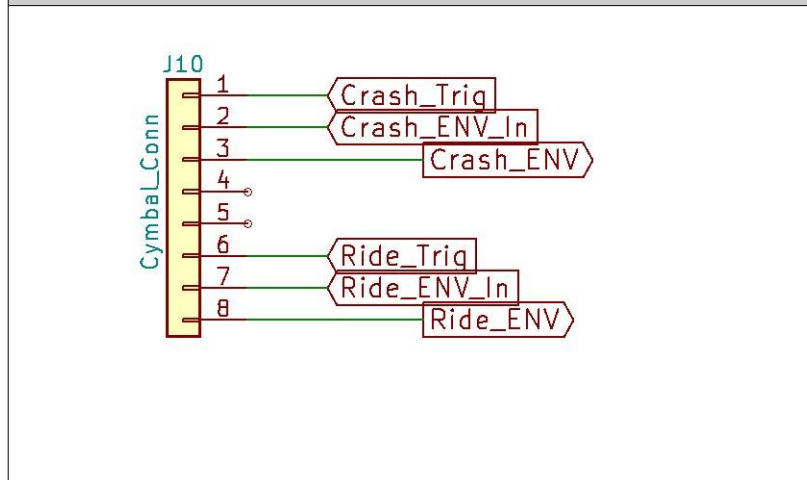
VR103	5k lin (trimmer)
VR113	5k lin (trimmer)

Install pots only after finishing all sections!

VR101	1M log (A1M)
VR102	500k log (A500k)
VR111	1M log (A1M)
VR112	500k log (A500k)

J10	1x8 pin header
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Cymbal - Connections



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Crash (alternative AD-ENV)

remove R432	
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```
[1] - Trig : C50
[2] - ENV_In : P433 (junction PA11x)
```

[3] - ENV : R432 (junction IC52B)

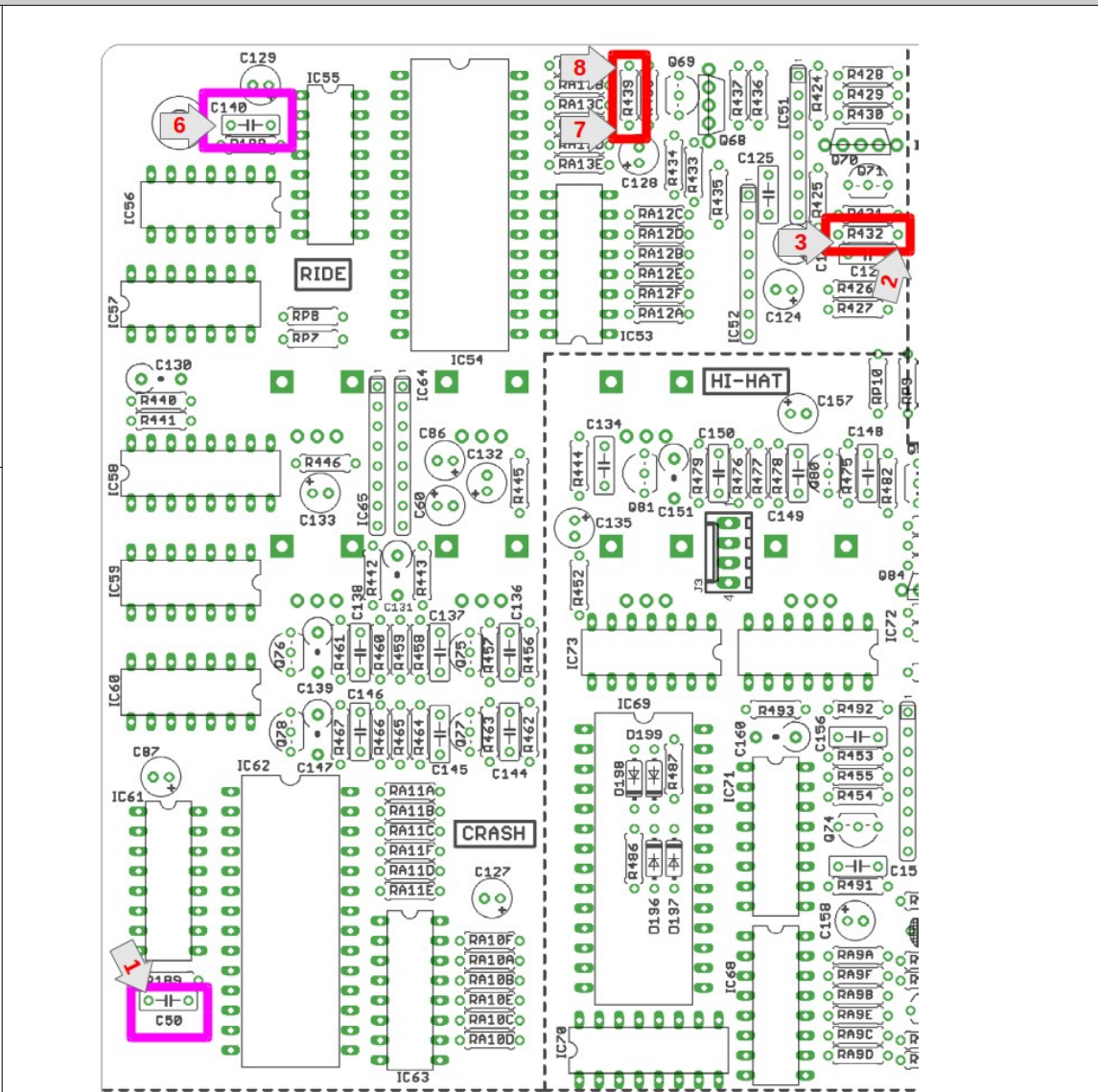
[3] J. L. Loefer, *Algebraic K-theory*, Graduate Texts in Mathematics, vol. 90, Springer-Verlag, New York, 1978.

Ride (alternative AD-ENV)

```
remove R439
[6] - Trig : C140
```

```
[7] - ENV In : R439 (junction RA13x)
```

[8] - ENV: R439 (junction IC52A)



Trimmer adjustments

VR15 - SnareDrum OSC2 Level

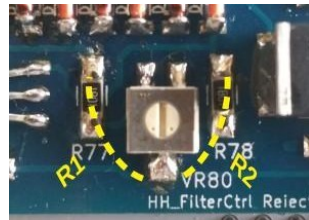
The additional Decay pot only affects one of two SD oscillators, OSC1. At slow decay values OSC1 appears much louder than OSC2, which is then almost inaudible. Use VR15 to increase OSC2 level to a more balanced/natural mix. The change is mainly noticeable at slow decay values but only slightly at shorts. Find your preferred spot or turn VR15 fully clockwise to cancel this mod.

VR80 - HiHat Filter-CV Reject

Short: Just set VR80 to middle position!

Long: It is in the nature of the Steiner-Parker Diode-VCF design that all CV signals are injected into the audio signal. To remove this unwanted effect the VCF uses a second equal filter path on which CV signals are injected phase inverted. By summing the two filtered signals the CV signal is canceled. With VR80 we adjust the balancing of pos. and neg. CV signals. Common method for Filter-CV rejection is to apply no audio signal to the filter but a LFO signal to the CV input at an audible frequency. Then adjust rejection pot until no signal is heard/measured at VCF output.

However, since we do not use a CV with an audible frequency in Extra9 HiHat VCF this procedure is not really needed. Instead, as we use matched transistors, we could ensure to have approximately similar resistances R1 and R2 as indicated on picture. It's expected to find this situation at center position of VR80.



VR85 - HiHat ENV-CV Bias

VR85 adjusts ENV CV bias. Measure voltage at pin3 of HH_ENV-Mod pot against GND and adjust VR85 until you read 0.0V

VR86 - Resonance Adjust

Use VR86 to adjust Filter max Resonance to your taste

VR103 / VR113 - Cymbal ENV Bias

Set ENV selector to AD, Attack to 0 and Decay to a low value (~10%). Now trigger the Cymbals. You may hear a residual sound after the shortened Cymbal sample. That is because the ENV does not close the amp cleanly. Adjust VR103 / VR113 until you don't hear such a residual anymore (should be around center position).